

LIARVATN, Norway

WMO: 014190

Lat: 59.0512N

Lon: 6.1213E

Elev: 984

StdP: 14.18

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.5	14.3	1.1	6.0	15.0	5.6	7.6	19.6	23.4	29.3	21.0	29.6	3.8	20	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	74.9	59.3	70.8	57.6	67.4	56.8	61.7	70.0	60.0	67.3	58.4	65.1	6.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.4	75.6	63.8	56.9	71.5	62.7	55.4	67.7	61.0	28.0	69.9	26.8	67.3	25.8	65.5	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.1	14.5	8.2	80.3	5.6	4.6	4.2	83.6	1.0	86.3	-2.1	88.8	-6.2	92.1
			7.3	63.8	5.5	2.4	3.3	65.5	0.1	66.9	-3.0	68.3	-7.0	70.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	31.5	31.7	34.9	41.3	48.6	53.2	57.5	56.4	52.4	45.2	38.6	33.9
	DBStd	11.12	6.64	7.11	5.85	6.16	7.44	4.49	5.83	4.21	4.79	4.93	6.43	7.91
	HDD50	2997	574	512	467	270	121	16	2	2	27	163	344	499
	HDD65	7746	1039	932	932	710	511	355	246	270	379	615	793	964
	CDD50	745	0	0	0	10	76	111	235	199	99	13	2	0
	CDD65	20	0	0	0	0	2	0	15	2	1	0	0	0
	CDH74	147	0	0	0	0	24	7	105	7	4	0	0	0
	CDH80	14	0	0	0	0	2	0	12	0	0	0	0	0
Wind	WSAvg	6.2	7.4	6.6	6.4	6.2	6.0	5.5	5.1	5.4	6.0	6.8	6.5	7.0
Precipitation	PrecAvg	69.4	7.3	6.1	5.4	3.7	3.2	3.7	4.6	5.8	6.9	8.0	7.6	7.8
	PrecMax	91.3	14.8	14.8	12.0	6.2	8.0	6.5	10.2	10.6	12.6	16.0	14.7	14.6
	PrecMin	47.4	0.0	0.0	0.2	1.6	0.2	0.7	2.0	1.4	2.1	2.1	0.9	0.7
	PrecStd	11.3	4.0	3.9	2.8	1.3	1.8	1.8	2.0	2.1	3.1	3.2	4.0	4.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	47.8	53.4	64.7	77.4	74.3	81.3	74.5	71.5	62.5	54.5	48.9
		MCWB	41.7	41.2	45.9	50.9	55.7	57.9	62.4	59.0	59.8	53.5	49.5	46.7
	2%	DB	43.1	43.6	48.4	61.0	71.5	68.6	77.0	69.2	65.5	58.1	51.5	46.6
		MCWB	40.9	39.9	43.5	49.3	55.3	56.1	60.6	57.3	56.3	50.9	47.3	44.7
	5%	DB	41.8	41.9	44.2	56.9	66.6	65.0	72.4	66.4	62.8	54.9	48.8	45.1
		MCWB	39.5	39.0	40.5	47.0	53.2	54.5	59.6	57.6	55.1	50.8	45.8	43.2
	10%	DB	40.2	40.3	42.3	52.2	61.9	61.7	68.3	64.1	60.3	52.2	46.9	43.3
		MCWB	37.9	37.8	39.4	43.0	51.5	53.0	58.7	55.8	54.0	47.6	44.0	41.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	42.5	47.4	55.4	59.7	61.4	64.7	62.0	61.1	55.7	50.3	47.5
		MCDB	44.2	44.4	51.7	62.0	72.0	69.2	76.4	68.0	69.6	59.3	52.9	48.0
	2%	WB	41.7	40.9	43.9	50.4	57.1	58.4	62.6	59.9	58.6	53.4	48.3	45.5
		MCDB	42.7	42.7	46.8	59.1	68.7	64.4	72.1	65.3	62.8	56.0	50.2	46.3
	5%	WB	40.2	39.5	41.9	47.0	54.5	56.1	61.0	58.3	56.7	51.1	46.5	43.4
		MCDB	41.7	41.7	43.7	55.0	63.0	62.7	69.3	64.2	61.3	54.1	48.6	44.8
	10%	WB	38.1	38.0	40.0	44.2	52.4	54.4	59.3	56.9	54.9	48.5	44.5	41.4
		MCDB	39.8	40.3	41.9	50.0	60.7	60.1	66.9	62.9	59.1	51.4	46.4	43.2
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.7	10.5	13.2	14.3	12.9	12.5	11.4	10.5	9.7	7.8	8.2
		MCDBR	8.2	9.9	13.7	21.4	22.3	19.5	20.7	14.7	15.5	13.4	9.4	7.9
		MCWBR	7.5	8.0	9.4	13.9	9.6	9.6	8.7	7.4	8.4	8.7	7.6	8.0
	5% WB	MCDBR	7.7	9.4	12.1	19.7	19.7	16.8	17.9	13.0	14.3	10.8	9.0	7.8
		MCWBR	7.7	8.1	9.4	14.5	9.8	9.8	8.5	7.4	8.7	8.3	8.1	8.1
Clear-Sky Solar Irradiance	taub	0.266	0.296	0.320	0.354	0.350	0.344	0.358	0.356	0.335	0.317	0.280	0.245	
	taud	2.171	2.271	2.291	2.294	2.430	2.477	2.453	2.455	2.511	2.469	2.281	2.103	
	Ebn at Noon	172	230	259	269	280	283	275	267	255	224	168	134	
	Edh at Noon	15	22	29	35	32	31	31	29	24	18	13	10	
All-Sky Solar Radiation	RadAvg	100	294	695	1226	1569	1710	1519	1173	754	365	134	65	
	RadStd	19	59	105	165	214	193	176	102	75	56	23	9	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-209	N/A	N/A

Nomenclature: See separate page